

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 03)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha ⁻¹	Carbon in soil dead organic matter / t ha ⁻¹
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 1

. 1

Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer

t/ha

0 1

. 2

Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 1

. 3

Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

1

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2

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5

0 2 . 1

Which of the following best describes a **representative** sample? Tick one box.**[1 mark]**

A A sample taken from the most convenient location

☐

B A sub-sample that accurately reflects the complete data set

☐

C The largest sample that can be collected

☐

D A sample that supports the hypothesis

☐

0 2 . 2

Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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0 2 . 3

Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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5

0 3 . 1

Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

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[illegible]

03 . 2

State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

10

04 . 1

Define the term **persistence** as applied to a pollutant.

[1 mark]

04 . 2

Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

5

0 5 Table 1 shows four general approaches to pollution control.

0 5 . **1** Complete Table 1 by giving one example of each approach.

[4 marks]

Table 1

Approach	Example
Production prevention	
Prevention of release	
Post-release remediation	
Alternative processes	

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0 5 . **2** Explain why prevention of production is generally preferable to post-release remediation.

[2 marks]

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0 5 . **3** Describe how each of the following new technologies helps to control pollution.

[4 marks]

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Satellite monitoring of oil spills

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GPS ship tracking

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Polymer adsorption of heavy metals

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Phytoremediation

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10

06 . 1

Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

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06 . 2

Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

Useful properties

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Properties causing ozone depletion

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0 6 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

1

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2

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3

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0 6 . 4 State what is meant by an **ozone hole**.

[1 mark]

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0 6 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

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06 . 6

Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

15

07 . 1

Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

07 . 2

Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

07 . 3

Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

07 . 4

Give **three** other positive feedback mechanisms.

[3 marks]

1

2

3

07 . 5

Explain what is meant by a climate **tipping point**.

[2 marks]

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15

08 . 1

Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐B Oxygen ☐C Argon ☐D Carbon dioxide ☐

08 . 2

Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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08 . 3

Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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5

09

Figure 1 is a flow diagram of the main stages in a sewage treatment works using activated sludge treatment.

Figure 1

Stage	Process
Pre-treatment	Screens and grit traps
Primary treatment	Primary sedimentation
Secondary treatment	Aeration tank then secondary sedimentation
Tertiary treatment	Phosphate stripping / micro-strainers / UV
Sludge treatment	Anaerobic digestion

09 . 1

Explain the purpose of screens and grit traps.

[2 marks]

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09 . 2

Explain why primary sedimentation is described as the most important process.

[2 marks]

09 . 3

Describe what happens in the aeration tank and explain why the sludge from the secondary sedimentation tank is returned to it.

[3 marks]

09 . 4

Explain why tertiary treatment may be needed before effluent is discharged, and describe **one** tertiary process.

[3 marks]

10

1 0

Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m ⁻² day ⁻¹	Mean solar energy in December / kWh m ⁻² day ⁻¹
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

1 0

. 1

Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

Answer

times

1 0

. 2

Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

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1 0

. 3

Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

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1 0

. 4

Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

10 . 5

Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

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Extra space

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15

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

1 1 . 2

Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a full page of a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

[illegible]

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS